EXPANSION VALVE

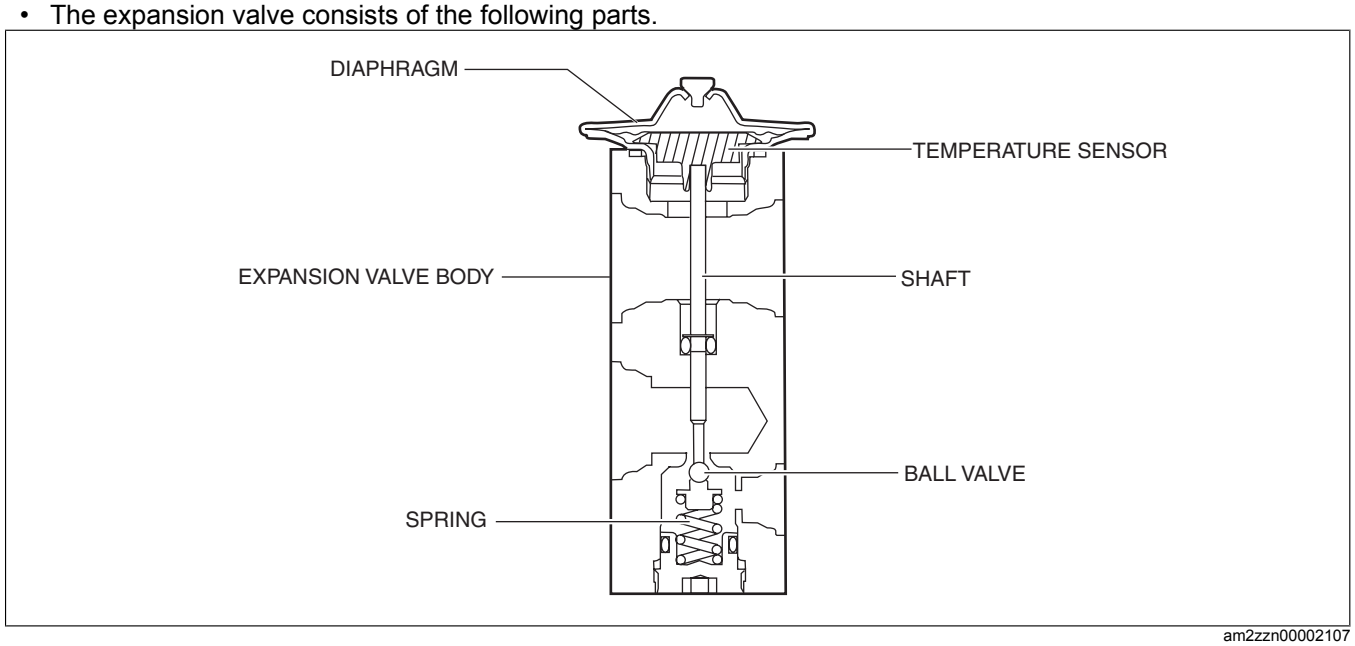
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Purpose, Function

- The expansion valve rapidly depressurizes the liquid refrigerant to form it into a mist.
- The expansion valve adjusts the liquid refrigerant flow volume sent to the evaporator.

Construction

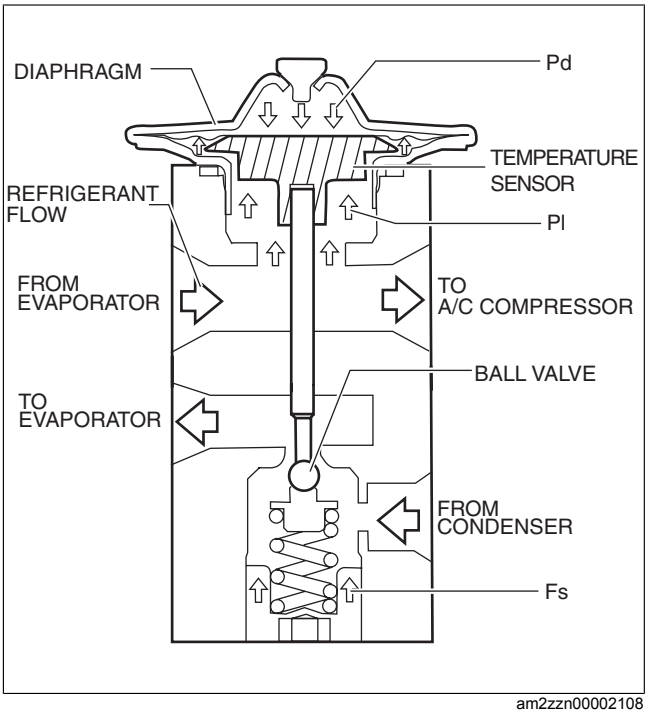
- The expansion valve consists of the following parts.



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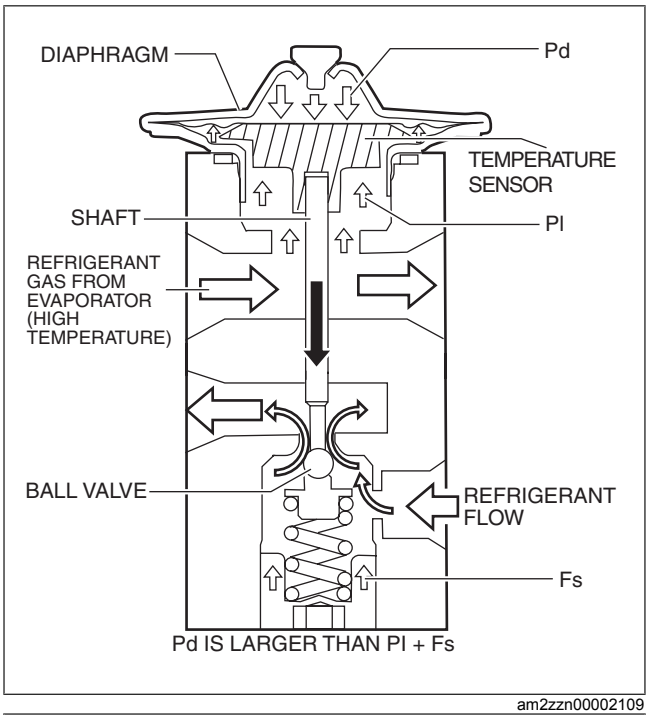
Operation

- The expansion valve adjusts the liquid refrigerant flow volume using the opening angle of the ball valve in the expansion valve.
- The opening angle adjustment is performed according to the balance of the following forces:
  - — Refrigerant pressure (Pd) in diaphragm
  - — Refrigerant gas pressure (PI) of evaporator under diaphragm
  - — Spring force (Fs) pushing up ball valve



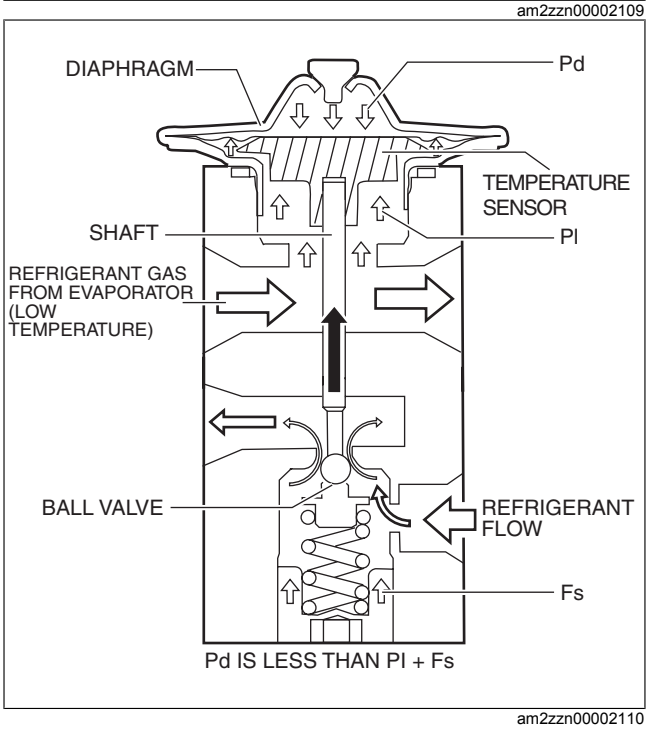
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- If the gaseous refrigerant temperature from the evaporator increases, the expansion valve increases the amount of refrigerant according to the following operations:
  1. The temperature of the temperature sensor tightly sealed against the diaphragm increases.
  2. The refrigerant in the diaphragm warms and the Pd increases.
  3. If this Pd increases more than PI + Fs, the diaphragm is pressed down.
  4. The shaft installed to the temperature sensor end presses down the ball valve.
  5. The amount of refrigerant is increased.



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- If the gaseous refrigerant temperature from the evaporator decreases, the expansion valve decreases the amount of refrigerant flow according to the following operation:
  1. The temperature of the temperature sensor tightly sealed against the diaphragm decreases.
  2. The refrigerant temperature in the diaphragm decreases and the Pd decreases.
  3. The Pd decreases less than PI + Fs.
  4. The ball valve installed to the temperature sensor end is pressed up.
  5. The amount of refrigerant is decreased.



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Fail-safe

- Not applicable